

# <演習解説> 8の字運動ノードを作ってみよう

## my\_node.cppの変更

```
#include "rclcpp/rclcpp.hpp"
#include "geometry_msgs/msg/twist.hpp"
#include "turtlesim/msg/pose.hpp"
#include <cmath>
#include <optional>

class EightFigureController : public rclcpp::Node
{
public:
    EightFigureController()
    : Node("eight_figure_controller"),
      linear_speed_(1.5),
      angular_speed_(1.0),
      current_angular_z_(angular_speed_),
      cumulative_angle_(0.0)
    {
        publisher_ = this->create_publisher<geometry_msgs::msg::Twist>("/turtle1/cmd_vel", 10);
        subscriber_ = this->create_subscription<turtlesim::msg::Pose>(
            "/turtle1/pose", 10,
            std::bind(&EightFigureController::pose_callback, this, std::placeholders::_1));
        timer_ = this->create_wall_timer(
            std::chrono::milliseconds(100),
            std::bind(&EightFigureController::timer_callback, this));
    }

private:
    void pose_callback(const turtlesim::msg::Pose::SharedPtr msg)
    {
        float theta = msg->theta;

        if (last_theta_) {
            float delta = theta - *last_theta_;

            // [-π, π] に正規化
            delta = std::fmod(delta + M_PI, 2 * M_PI);
            if (delta < 0) delta += 2 * M_PI;
            delta -= M_PI;

            cumulative_angle_ += delta;

            if (std::abs(cumulative_angle_) > 2 * M_PI) {
                current_angular_z_ *= -1.0;
                cumulative_angle_ = 0.0;
                RCLCPP_INFO(this->get_logger(), "→ 方向反転");
            }
        }

        last_theta_ = theta;
    }
};
```

```

}

void timer_callback()
{
    geometry_msgs::msg::Twist msg;
    msg.linear.x = linear_speed_;
    msg.angular.z = current_angular_z_;
    publisher_→publish(msg);
}

rclcpp::Publisher<geometry_msgs::msg::Twist>::SharedPtr publisher_;
rclcpp::Subscription<turtlesim::msg::Pose>::SharedPtr subscriber_;
rclcpp::TimerBase::SharedPtr timer_;
float linear_speed_;
float angular_speed_;
float current_angular_z_;
double cumulative_angle_;
std::optional<float> last_theta_;
};

int main(int argc, char * argv[])
{
    rclcpp::init(argc, argv);
    rclcpp::spin(std::make_shared<EightFigureController>());
    rclcpp::shutdown();
    return 0;
}

```

## CMakeLists.txtの修正

```

# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(geometry_msgs REQUIRED)
find_package(turtlesim REQUIRED) # ← これを追加

ament_target_dependencies(my_cpp_node
    rclcpp
    std_msgs
    geometry_msgs
    turtlesim # ← これを追加
)

```

## パッケージのビルド

```

cd ~/colcon_ws
colcon build
source install/setup.bash

```

## 別ターミナルでTurtlesimを起動

```

ros2 run turtlesim turtlesim_node

```

## 元ターミナルで自作ノードを起動

```
ros2 run my_py_pkg my_node
```